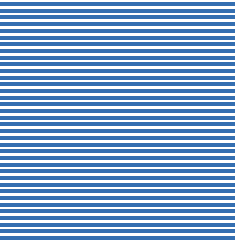


Architecture shape and parameter budgets
Qwen3-8B Dense variants | Qwen3-235B expert variants

Dense | baseline (released baseline)



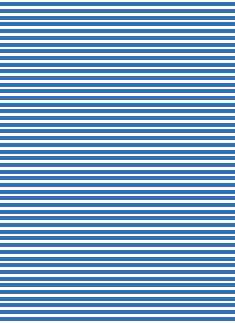
L=36; H=4,096; F=12,288

gate/up: [12,288, 4,096] each
down: [4,096, 12,288]

Total parameters: 8,190,735,360
Delta: +0

One bar per layer; bar width proportional to H

Dense | deeper same width (untrained)



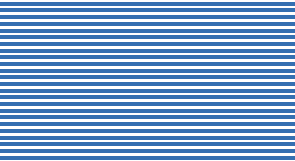
L=48; H=4,096; F=8,320

gate/up: [8,320, 4,096] each
down: [4,096, 8,320]

Total parameters: 8,165,670,912
Delta: -25,064,448

One bar per layer; bar width proportional to H

Dense | shallower wider (untrained)



L=24; H=5,120; F=13,952

gate/up: [13,952, 5,120] each
down: [5,120, 13,952]

Total parameters: 8,209,296,384
Delta: +18,561,024

One bar per layer; bar width proportional to H

MoE | baseline (released baseline)



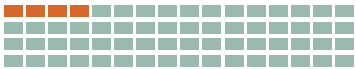
L=94; E=128; top-k=8
H=4,096; F=1,536

per expert gate/up: [1,536, 4,096]
down: [4,096, 1,536]

Total: 235,093,634,560
Delta: +0
Router: 49,283,072

One layer of experts; orange = first synthetic route top-k

MoE | coarse64 (untrained)



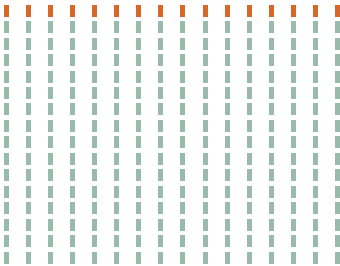
L=94; E=64; top-k=4
H=4,096; F=3,072

per expert gate/up: [3,072, 4,096]
down: [4,096, 3,072]

Total: 235,068,993,024
Delta: -24,641,536
Router: 24,641,536

One layer of experts; orange = first synthetic route top-k

MoE | fine256-k16 (untrained)



L=94; E=256; top-k=16
H=4,096; F=768

per expert gate/up: [768, 4,096]
down: [4,096, 768]

Total: 235,142,917,632
Delta: +49,283,072
Router: 98,566,144

One layer of experts; orange = first synthetic route top-k

Symbolic scales differ between columns. FFN weights shown, not all model matrices. No equal-quality or speed claim.